

Project Design Phase

Proposed Solution Template

Date	JUNE 2026
Team ID	
Project Name	Plugging-into-the-Future-An-Exploration-of-Electricity-Consumption-Patterns
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Electricity consumption datasets are an important resource for understanding energy usage patterns, yet they are presented in raw, tabular form that is difficult to interpret and explore visually. Researchers, policy makers, educators, and energy analysts often lack the tools to derive meaningful insights about electricity consumption trends, regional usage, seasonal variations, and consumption patterns. This limits understanding and informed decision-making for energy planning, conservation, and sustainability.
2.	Idea / Solution description	We propose the development of a comprehensive, interactive Tableau dashboard that visualizes the Electricity Consumption dataset. The dashboard will display electricity consumption by region, time period, sector, and usage patterns. It will include maps, bar charts, line charts, timelines, and filters to enable users to explore insights at regional, national, or sectoral levels. The solution will be hosted on Tableau Public for open access.
3.	Novelty / Uniqueness	Unlike existing static reports or basic charts, our solution provides live, filterable, and comparative visualizations that combine geographic and temporal insights. The use of Tableau allows for real-time exploration without the need for coding, making the solution accessible to both technical and non-technical users in the energy sector. No official interactive visualization tool of this depth exists for this dataset currently.

4.	Social Impact / Customer Satisfaction	This project enhances awareness of electricity consumption patterns and supports informed decision-making for energy management and sustainability. Policy makers, energy planners, educators, researchers, and students will benefit from easy access to visual insights. The platform enables deeper understanding of regional consumption differences, usage trends, and long-term energy patterns—thereby supporting more efficient energy planning and conservation efforts.
5.	Business Model (Revenue Model)	This is a non-commercial, open-access project primarily intended for social good and awareness. However, in the long term, this model could support: • Partnerships with energy organizations, utility companies, or government bodies • Workshops/training programs on energy data visualization • Customized dashboards for region-specific or research-specific use cases on a paid or grant-supported basis
6.	Scalability of the Solution	The platform is highly scalable. It can: • Be updated with newer Electricity Consumption datasets periodically • Be expanded to include renewable energy analysis, power generation statistics, or energy demand forecasting • Be translated and localized for global use • Integrate real-time APIs or mobile dashboards in the future